

MAFGA Multi-Agent Architecture Diagram

This document contains the architecture diagram for the Multi-Agent POC built on MAF 1.0 GA.

High-Level Architecture

```
flowchart TB
    subgraph Client["Client Layer"]
        UI[Web UI / CLI]
        API_Client[API Client]
    end

    subgraph Gateway["API Gateway"]
        FastAPI[FastAPI Service]
        Auth[EntraID Auth]
        CORS[CORS Middleware]
    end

    subgraph Middleware["Middleware Pipeline"]
        RBAC[RBAC Middleware]
        ContentSafety[Content Safety]
        AuditLog[Audit Log]
    end

    subgraph Orchestration["Orchestration Layer"]
        Magentic[Magentic Orchestrator<br/>Task-Ledger Planning]
        Router[Traffic Router<br/>Intent-Based Routing]
        Parallel[Parallel Executor<br/>Concurrent Execution]
        HITL[Human Approval<br/>Manager]
    end

    subgraph Agents["Agent Layer"]
        MerchPlanner[MerchPlanner Agent]
        SpacePlanner[SpacePlanner Agent]
        LoyaltyAgent[LoyaltyAgent]
        ProductsFinder[ProductsFinder Agent]
        CommercialSales[CommercialSales Agent]
        CampaignAnalyst[CampaignAnalyst Agent]
    end

    subgraph Memory["Memory & State"]
        Redis[(Redis Memory)]
        FileCheckpoint[(File Checkpoint)]
        InMemory[(In-Memory Cache)]
    end

    subgraph MCP["MCP Servers (Tools)"]
        Snowflake[Snowflake MCP<br/>Data Warehouse]
        Salesforce[Salesforce MCP<br/>CRM]
```

```

    Weather[Weather MCP<br/>Forecasts]
    Items[Items API MCP<br/>Product Catalog]
    Personal[Personalisation MCP<br/>Preferences]
    Local[Localisation MCP<br/>Regional Data]
end

subgraph A2A["A2A External Agents"]
    PricingAgent[Pricing Agent]
    InventoryAgent[Inventory Agent]
    FulfilmentAgent[Fulfilment Agent]
end

subgraph LLM["LLM Providers"]
    AzureFoundry[Azure AI Foundry]
    AzureOpenAI[Azure OpenAI]
    OpenAI[OpenAI]
    Anthropic[Anthropic Claude]
    Ollama[Ollama Local]
end

subgraph Observability["Observability"]
    Telemetry[OpenTelemetry]
    Jaeger[Jaeger Tracing]
    Prometheus[Prometheus Metrics]
end

%% Client connections
UI --> FastAPI
API_Client --> FastAPI

%% Gateway flow
FastAPI --> Auth
Auth --> CORS
CORS --> RBAC

%% Middleware chain
RBAC --> ContentSafety
ContentSafety --> AuditLog

%% Middleware to Orchestration
AuditLog --> Magentic
AuditLog --> Router
AuditLog --> Parallel

%% Orchestration dependencies
Magentic --> HITL
Magentic --> MerchPlanner
Magentic --> SpacePlanner
Magentic --> LoyaltyAgent
Magentic --> ProductsFinder
Magentic --> CommercialSales
Magentic --> CampaignAnalyst

```

```
Router --> MerchPlanner
Router --> SpacePlanner
Router --> LoyaltyAgent
Router --> ProductsFinder
Router --> CommercialSales
Router --> CampaignAnalyst
```

```
Parallel --> MerchPlanner
Parallel --> SpacePlanner
```

```
%% Agent to MCP
```

```
MerchPlanner --> Snowflake
SpacePlanner --> Snowflake
LoyaltyAgent --> Personal
LoyaltyAgent --> Local
ProductsFinder --> Items
CommercialSales --> Salesforce
CommercialSales --> Snowflake
CampaignAnalyst --> Weather
CampaignAnalyst --> Snowflake
```

```
%% Agent to A2A
```

```
MerchPlanner --> PricingAgent
ProductsFinder --> InventoryAgent
CommercialSales --> FulfilmentAgent
```

```
%% Agent to LLM (via factory)
```

```
MerchPlanner -.-> AzureFoundry
SpacePlanner -.-> AzureFoundry
LoyaltyAgent -.-> AzureFoundry
ProductsFinder -.-> AzureFoundry
CommercialSales -.-> AzureFoundry
CampaignAnalyst -.-> AzureFoundry
```

```
%% Memory connections
```

```
MerchPlanner --> Redis
MerchPlanner --> FileCheckpoint
```

```
%% Observability
```

```
FastAPI --> Telemetry
Magentic --> Telemetry
MerchPlanner --> Telemetry
Telemetry --> Jaeger
Telemetry --> Prometheus
```

MAF 1.0 GA Component Detail

```
flowchart LR
```

```
    subgraph AgentPattern["Agent Pattern (MAF 1.0 GA)"]
        Agent[Agent Class]
        ChatClient[ChatClient Interface]
        Instructions[System Instructions]
```

```

    Tools[Tool Definitions]
end

subgraph ChatClients["Pluggable Chat Clients"]
    AzureFoundryClient[AzureAI<br/>ChatCompletions<br/>Client]
    AzureOpenAIClient[AzureOpenAI<br/>Client]
    OpenAIClient[OpenAI<br/>Client]
    AnthropicClient[Anthropic<br/>Client]
    OllamaClient[Ollama<br/>Client]
end

subgraph MCPIntegration["MCP Tool Integration"]
    MCPToolCall[Tool Call Request]
    MCPServer[MCP Server]
    MCPResult[Tool Result]
end

Agent --> ChatClient
Agent --> Instructions
Agent --> Tools

ChatClient --> AzureFoundryClient
ChatClient --> AzureOpenAIClient
ChatClient --> OpenAIClient
ChatClient --> AnthropicClient
ChatClient --> OllamaClient

Tools --> MCPToolCall
MCPToolCall --> MCPServer
MCPServer --> MCPResult
MCPResult --> Agent

```

Orchestration Patterns

```

flowchart TB
    subgraph MagenticPattern["Magentic Orchestration"]
        direction TB
        Goal[User Goal]
        Planner[LLM Planner]
        Plan[Task Ledger / Plan]
        Approval{Human<br/>Approval?}
        ExecuteSteps[Execute Steps]
        Synthesize[Synthesize Results]

        Goal --> Planner
        Planner --> Plan
        Plan --> Approval
        Approval -->|Approved| ExecuteSteps
        Approval -->|Rejected| Goal
        ExecuteSteps --> Synthesize
    end
end

```

```

subgraph HandoffPattern["Handoff (Traffic Routing)"]
    direction TB
    Query[User Query]
    Classifier[Intent Classifier]
    RouteDecision{Select Agent}
    Agent1[Agent A]
    Agent2[Agent B]
    Agent3[Agent C]

    Query --> Classifier
    Classifier --> RouteDecision
    RouteDecision --> Agent1
    RouteDecision --> Agent2
    RouteDecision --> Agent3
end

subgraph ConcurrentPattern["Concurrent Execution"]
    direction TB
    Task[Multi-task Request]
    Split[Split Tasks]
    ParallelExec["Parallel Execution<br/>(asyncio.gather)"]
    Aggregate[Aggregate Results]

    Task --> Split
    Split --> ParallelExec
    ParallelExec --> Aggregate
end

```

Human-in-the-Loop Workflow

```

sequenceDiagram
    participant User
    participant API
    participant Orchestrator
    participant ApprovalManager
    participant Agent

    User->>API: POST /orchestration/run
    API->>Orchestrator: run(goal, require_approval=true)
    Orchestrator->>Orchestrator: Create Plan
    Orchestrator->>ApprovalManager: Request Approval
    ApprovalManager-->>API: Return pending approval
    API-->>User: Plan pending approval

    Note over User,ApprovalManager: Human reviews plan

    User->>API: POST /orchestration/approvals/{id}
    API->>ApprovalManager: approve(id)
    ApprovalManager->>Orchestrator: Approval granted
    Orchestrator->>Agent: Execute step 1
    Agent-->>Orchestrator: Result 1
    Orchestrator->>Agent: Execute step 2

```

```
Agent-->>Orchestrator: Result 2
Orchestrator-->>Orchestrator: Synthesize results
Orchestrator-->>API: Final result
API-->>User: Orchestration complete
```

Memory & Checkpoint Flow

```
flowchart LR
    subgraph Conversation["Conversation Flow"]
        Query1[Query 1]
        Query2[Query 2]
        Query3[Query 3]
    end

    subgraph MemorySystem["Memory System"]
        MemoryProvider[Memory Provider]
        Redis[(Redis)]
        InMem[(In-Memory)]
    end

    subgraph CheckpointSystem["Checkpoint System"]
        CheckpointStore[Checkpoint Store]
        FileStore[(File Store)]
    end

    Query1 -->|Store| MemoryProvider
    Query2 -->|Store| MemoryProvider
    Query3 -->|Store| MemoryProvider

    MemoryProvider --> Redis
    MemoryProvider --> InMem

    MemoryProvider -->|Periodic Save| CheckpointStore
    CheckpointStore --> FileStore

    FileStore -->|Restore on Restart| MemoryProvider
```

Security Architecture

```
flowchart TB
    subgraph AuthFlow["Authentication Flow"]
        Request[Incoming Request]
        JWT[JWT Token]
        EntraID[Azure EntraID]
        UserContext[User Context]
    end

    subgraph AuthZ["Authorization"]
        RBAC[RBAC Check]
        AgentAccess{Agent<br/>Access?}
        Allow[Allow]
        Deny[Deny 403]
    end
```

end

```
subgraph Safety["Content Safety"]
  PIICheck[PII Detection]
  KeywordFilter[Keyword Filter]
  ContentMod[Content Moderation]
end
```

```
subgraph Audit["Audit Trail"]
  AuditLog[(Audit Log)]
  Compliance[Compliance Report]
end
```

```
Request --> JWT
JWT --> EntraID
EntraID --> UserContext
UserContext --> RBAC
RBAC --> AgentAccess
AgentAccess -->|Yes| PIICheck
AgentAccess -->|No| Deny
PIICheck --> KeywordFilter
KeywordFilter --> ContentMod
ContentMod --> Allow
```

```
Allow --> AuditLog
Deny --> AuditLog
AuditLog --> Compliance
```